Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Cabrera, Marco E. Ph.D.		
Project Title:	Metabolic Adaptation of Skeletal Muscle to Training/Detraining: A Systems Model		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Nutrition, Physical Fitness, and Rehabilitation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cleveland	**State:**	OH
Zip Code:	44106-6011	**Congressional District:**	11
Comments:	NOTE: PI passed away in early February 2009.		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	05/01/2001	**End Date:**	09/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: project extended to 9/30/05		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Radhakrishnan, Krishnan (Ohio Aerospace Institute) Saidel, Gerald (Case Western Reserve University) Stanley, William (Case Western Reserve University)		
Grant/Contract No.:	NCC 9-58-NPFR00206		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space travel (detraining) has detrimental effects on skeletal muscle structure, metabolism, and function, including reductions in muscle size, strength, and endurance. Exercise (training) in space can counteract some of these deleterious effects. Indeed, experimental studies are still being conducted to determine both the cause of muscle deterioration and the type and characteristics of exercise training programs needed to counteract muscle atrophy. Physiologically-based computational models of human function can integrate cellular to whole-body data, provide a framework for quantitative understanding of the skeletal muscle metabolic responses to exercise after loading or unloading, and help in the development of countermeasures. (1) The original specific aims of this project were: a) To identify the metabolic adaptations to training and detraining to develop databases containing information on the structural, metabolic, and functional adaptations of skeletal muscle to microgravity and exercise training and the underlying biochemical mechanisms mediating these adaptations. b) To develop mathematical models of intermediary metabolism in skeletal muscle that account for the effects of training and detraining. c) To investigate the relative significance of model parameters affected by training or detraining on work capacity and efficiency. d) To simulate the effects on skeletal muscle intermediary metabolism and energetics of space flight and exercise in space, to quantitatively test the hypotheses that after a period of space travel or exercise training, the observed changes in the rates of carbohydrate and fatty acid oxidation in skeletal muscle are a result of a partial conversion of slow-twitch to fast-twitch fibers and alterations in glycolytic and oxidative enzymes. (2) Key findings of the project: from model simulations we were able to link changes in enzyme activities at the cellular level to patterns of fuel utilization at the whole tissue level. (3) This finding has great potential for the optimazation of training regimes as countermeasures. Once the model of the whole body is completed the effects of dietary manipulations could be investigated. During the third year of the award we concentrated on the development and implementation of the computational model of whole-body metabolism. This model integrates cellular to whole-body data and incorporates parameters representing enzyme activities which have been identified to be altered with chronic loading or unloading. (4) The mathematical model of the whole body is currently being coded and we expect to have it completed over the next six months to perform computer simulations of physiological responses to exercise, dietary changes, and/or environmental conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Computer model developed can be extended to predict effects of aging or to investigate hypotheses on the mechanisms leading to increase insulin resistance with bed rest
Task Progress:	1. Conducted and analyzed simulations of the effect of changes in model parameters representing enzyme activities on muscle metabolic responses during exercise. (JAD) 2. Developed, validated, and tested the robustness of model of cardiac metabolism and validated model with experimental data. (JES) 3. Compiled information about metabolic characteristics of brain, liver and other tissues to develop a database of metabolic information, which can be used for the development of a computational model of whole-body metabolism that integrates cellular information. (JES) 4. Established the modeling framework and metabolic pathways required for developing the model of whole body using a top-down approach. Whole-body model is being developed using less complex modules for brain, liver, and other tissues compartments, while keeping the complexity of the models of skeletal and cardiac muscle metabolism developed during the course of the award.
Bibliography Type:	Description: (Last Updated: 08/15/2018)
Articles in Peer-reviewed Journals	Salem JE, Stanley WC, Cabrera ME. "Computational studies of the effects of myocardial blood flow reduction on cardiac metabolism. " Biomed Eng Online. 2004 Jun 2;3(1):15. PMID: 15175110 , Jun-2004
Articles in Peer-reviewed Journals	Salem JE, Cabrera ME, Chandler MP, Mc Elfresh TA, Huang H, Sterk JP, Stanley WC. "Step and ramp induction of myocardial ischemia: comparison of in vivo and in silico results." J Physiol Pharmacol. 2004 Sep;55(3):519-36. PMID: 15381824 , Sep-2004
Presentation	Salem JE, Cabrera ME, Chandler MP, Mc Elfresh TA, Huang H, Sterk JP, and Stanley WC. "Step and ramp induction of myocardial ischemia: comparison of in vivo and in silico results. " Sep-2003
Presentation	Cabrera ME, Di Bella JA, Saidel GM, Radhakrishnan K, Stanley WC. "A Mathematical Model of Skeletal Muscle Metabolism: Simulations of exercise responses after space travel and endurance training " Oct-2004
Presentation	Di Bella JA, Radhakrishnan K, and Cabrera ME "Metabolic Adaptations to Training/Detraining. A Systems Model " May-2004